

Evolutionary computing

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Experimental Framework

Exp ID	Research Question	Methodology	Input Parameters	Success Metrics	Expected Outcome
EXP-1	Can we reproduce the 2020 paper's null result?	Implement exact 2020 model. Use GA + ANN.	14 flux integers, 6 moduli fields	% critical points, Stability ratio	Confirm absence of stable dS vacua
EXP-2	Does torsion geometry enable stable dS states?	Implement 2022 model. Use Hybrid SA+CG.	5 flux coefficients, 2 moduli	Number of stable dS, Flatness measure	Verify claim of 170 stable dS vacua
EXP-3	Which ML strategy finds viable vacua fastest?	Compare: GA, SA, NN, PSO on both models	Identical parameter spaces	Convergence speed, Solution quality	Identify optimal algorithm
EXP-4	How sensitive are "stable" dS vacua?	Apply perturbations: quantum corrections, flux variations	Perturbation strength parameters	Lifetime estimation, Decay channels	Classify stability vs metastability
EXP-5	Where are Swampland boundaries?	Systematic scan near conjectured boundaries	c' parameter, moduli mass ratios	Distribution in (m ² /V) space	Precisely map Swampland boundary
EXP-6	How do results scale with complexity?	Gradually increase: moduli, fluxes, geometries	Dimensionality parameters	Computational scaling laws	Understand realistic model limitations

Research Hypothesis & Novel Contributions

Hypothesis

The claimed stable de Sitter vacua in torsional compactifications are actually **metastable** when subjected to robust stability analysis, and advanced ML algorithms can precisely quantify their lifetime and decay channels.

Novel Contributions

- First systematic comparison of ML methods for string landscape exploration
- Quantitative stability analysis of claimed dS vacua
- Empirical mapping of Swampland conjecture boundaries
- Scalability analysis for computational string theory

Methodology Overview

Algorithms to Compare

- Genetic Algorithm (GA)
- Simulated Annealing (SA)
- Neural Networks (NN)
- Particle Swarm (PSO)
- Hybrid SA+CG

Evaluation Metrics

- Convergence speed
- Solution quality ($V_{\min}$)
- Computational cost
- Stability classification
- Boundary precision

Technical Implementation

- **Python** (TensorFlow/PyTorch for ML)
- High-performance computing cluster
- Parallel evaluation of flux configurations
- Automated stability analysis pipeline

Expected Deliverables

- Comparative analysis framework
- Stability classification system
- Boundary mapping methodology
- Open-source code repository
- Thesis document

Success Criteria

- Reproduce published results
- Identify optimal algorithms
- Quantify stability bounds
- Provide scalable methodology

References I

- [1] Nana Cabo Bizet et al. "Testing swampland conjectures with machine learning." In: *The European Physical Journal C* 80.8 (Aug. 2020). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-020-8332-9](https://doi.org/10.1140/epjc/s10052-020-8332-9). URL: <http://dx.doi.org/10.1140/epjc/s10052-020-8332-9>.
- [2] Cesar Damian and Oscar Loaiza-Brito. "Galois groups of uplifted de Sitter vacua." In: *Annals of Physics* 467 (Aug. 2024), p. 169697. ISSN: 0003-4916. DOI: [10.1016/j.aop.2024.169697](https://doi.org/10.1016/j.aop.2024.169697). URL: <http://dx.doi.org/10.1016/j.aop.2024.169697>.
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